

Recitation 9: Solow Growth Model

Consider an economy described by the Solow growth model where the production function is given by $Y = K^{0.5}L^{0.5}$, where K is the capital stock and L is labor, which remains constant over time. Assume that the savings rate (s) is 0.25, the depreciation rate (δ) is 0.05, and the labor force (L) is normalized to 1.

1. Calculate the steady-state level of capital (K^*) and output (Y^*).
2. Suppose the savings rate increases to 0.35. What will be the new steady-state level of capital and output?
3. Discuss the transition dynamics. How does the economy transition from the old steady-state to the new one after the increase in the savings rate?